

5-Mark Questions & Answers -- DBMS Module 2: Relational Algebra & Calculus

Q1. Explain all fundamental operations of Relational Algebra with notation and examples.

1. Selection (sigma): $\sigma_{\text{Dept_ID} = 10}(\text{Employee})$ -- Selects rows where Dept_ID is 10.

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Q2. Explain Outer Join operations (Left, Right, Full) with concrete table diagrams.

- * Left Outer Join ($\bowtie_{\text{L}}$): Retains all tuples from the left table; unmatched right attributes are filled with NULL.

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Q3. Explain Domain Relational Calculus (DRC) and compare TRC vs DRC.

- * DRC Form: $\{ \langle x_1, x_2, \dots, x_n \rangle \mid P(x_1, x_2, \dots, x_n) \}$ where variables represent attribute domains rather than full tuples.

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